Question:

John drives for 3 hours at a speed of 60 mph and then turns around because he realizes he forgot something very important at home. He tries to get home in 4 hours but spends the first 2 hours in standstill traffic. He spends the next half-hour driving at a speed of 30mph, before being able to drive the remaining time of the 4 hours going at 80 mph. How far is he from home at the end of those 4 hours?

Reasoning + Answer:

Let's break down the problem step by step to determine how far John is from home at the end of the 4 hours of driving after turning around .

1 . ** Initial Driving to Real ization : **
- John drives for 3 hours at a speed of 6 0 mph .
- Distance covered initially :
\ [
\ text { Distance } = \ text { Speed } \ times \ text { Time } = 6 0 \ , \ text { mph } \ times 3 \ , \ text { hours } = 1 8 0 \ , \ text { m iles }
\ ]

2 . ** Driving Back wards to Turn Around : **
- John drives back 1 8 0 miles to get home .
- Since he covers 1 8 0 miles to return to the starting point :
\ [
\ text { Distance covered to turn around } = 1 8 0 \ , \ text { m iles }
\ ]

3 . ** First Leg of the Return Journey : **
- John spends 2 hours in stand still traffic .
- Distance covered in stand still traffic :
\ [
\ text { Distance } = 0 \ , \ text { mph } \ times 2 \ , \ text { hours } = 0 \ , \ text { m iles }
\ ]

4 . ** Second Leg of the Return Journey : **
- John then drives for 3 0 minutes (0 . 5 hours) at a speed of 3 0 mph .
- Distance covered in the first part of the return journey :
\ [
\ text { Distance } = 3 0 \ , \ text { mph } \ times 0 . 5 \ , \ text { hours } = 1 5 \ , \ text { m iles }
\ ]

5 . ** Final Leg of the Return Journey : **
- John has 1 . 5 hours left to cover the remaining distance at a speed of 8 0 mph .
- Let \ (d \ ) be the remaining distance to cover at 8 0 mph .
- Total distance to be covered is 1 8 0 miles (initial distance plus the return distance):
\ [
\ text { Total distance } = 1 8 0 \ , \ text { m iles }
\ ]
- Distance already covered :
\ [
\ text { Distance covered } = 0 \ , \ text { m iles } + 1 5 \ , \ text { m iles } = 1 5 \ , \ text { m iles }
\ ]
- Remaining distance to cover :
\ [
d = 1 8 0 \ , \ text { m iles } - 1 5 \ , \ text { m iles } = 1 6 5 \ , \ text { m iles }
\ ]
- Distance covered in the final leg :
\ [
\ text { Distance } = 8 0 \ , \ text { mph } \ times 1 . 5 \ , \ text { hours } = 1 2 0 \ , \ text { m iles }
\ ]

6 . ** Verification : **
- Check if the total distance covered matches the required 1 8 0 miles :
\ [
1 5 \ , \ text { m iles } + 1 2 0 \ , \ text { m iles } = 1 3 5 \ , \ text { m iles }
\ ]
- There seems to be an inconsistency as the total should be 1 8 0 miles . Re -e valuate the remaining distance :
\ [
\ text { Remaining distance } = 1 8 0 \ , \ text { m iles } - 1 5 \ , \ text { m iles } = 1 6 5 \ , \ text { m iles }
\ ]
- Correct ing the final leg :
\ [
\ text { Distance } = \ frac { 1 6 5 \ , \ text { m iles } } { 8 0 \ , \ text { mph } } \ times 1